

2nd International Conference on Sustainability, Innovation & Technology (ICSIT 2026)) : Submission (2200) has been created.

1 message

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Sat, May 9, 2026 at 12:08 PM

Hello,

The following submission has been created.

Track Name: ICSIT2026

Paper ID: 2200

Paper Title: Beyond Non-Maximum Suppression: A Comparative Latency and Accuracy Analysis of YOLO26 versus RT-DETR on the Symile-MIMIC Chest X-ray Dataset

Abstract:

Non-maximum suppression (NMS) is a post processing step in object detection to remove duplicate boxes. It retains the strongest predictions and removes the others but it also introduces delay and can eliminate medical findings (such as overlapping lesions). More recent models do not have NMS. They have end-to-end training where the model outputs the results. But there is a lack of literature comparing NMS-free CNN-based and Transformer-based detectors in medical imaging. This paper presents the performance comparison of CNN and YOLO and RT-DETR, for chest X-ray analysis on Symile-MIMIC dataset. It is composed of data cleansing, YOLO format converting and data augmentation. Performance measured using Mean Average Precision (mAP), recall and inference time. The results, with an mAP50 of 0.518 and an inference time of 13.62 ms for YOLO26 Nano, and an mAP50 of 0.275 with an inference time of 46.74 ms for RT-DETR Large, indicate the accuracy-speed trade-off in NMS-free medical object detection. CNNs are also fast but RT-DETR is better in capturing the global information with global attention. The findings highlight the importance of quality data and analysis, and demonstrate the potential of NMS-free detectors for real-time medical applications.

Created on: Sat, 09 May 2026 06:37:54 GMT

Last Modified: Sat, 09 May 2026 06:37:54 GMT

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Primary Subject Area: Track 2: Computational Intelligence & Machine Learning

Secondary Subject Areas: Not Entered

Submission Files:

DL_PAPER.pdf (626 Kb, Sat, 09 May 2026 06:33:49 GMT)

Submission Questions Response:

1. Originality Confirmation
Yes
2. AI Usage Disclosure
No
3. Keywords
NMS-Free Object Detection, YOLO26, RT DETR,
Medical Imaging, Chest X-ray Analysis, Inference Latency
4. Contact
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5. Contact
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6. Author Consent
Agreement accepted
7. Format Compliance
Agreement accepted
8. Author Limit Policy (6 Authors)
Agreement accepted

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